

Lab 2

Introduction to Databases



SQL (Structured Query Language)

SQL, or **Structured Query Language**, is the standard programming language used to **manage and manipulate relational databases**. It allows you to perform operations such as querying data, updating records, creating or modifying database structures, and controlling access to data.

Basic Concepts

Database: A structured collection of data stored electronically.

Table: A collection of rows and columns within a database.

Row (Record): A single entry in a table.

Column (Field): A single attribute or property of data.

Primary Key: A unique identifier for each row in a table.

Foreign Key: A column that creates a link between two tables.

Naming Rules

- Table names and column names:
 - Must begin with a letter
 - Must be 1–30 characters long
 - Must contain only A–Z, a–z, 0–9, _, \$, and #
 - Must not duplicate the name of another object owned by the same user
 - Must not be an Oracle server reserved word

Data Types

Data Type	Description
VARCHAR2 (<i>size</i>)	Variable-length character data
CHAR (<i>size</i>)	Fixed-length character data
NUMBER (<i>p</i> , <i>s</i>)	Variable-length numeric data
DATE	Date and time values
LONG	Variable-length character data (up to 2 GB)
CLOB	Character data (up to 4 GB)
RAW and LONG RAW	Raw binary data
BLOB	Binary data (up to 4 GB)
BFILE	Binary data stored in an external file (up to 4 GB)
ROWID	A base-64 number system representing the unique address of a row in its table

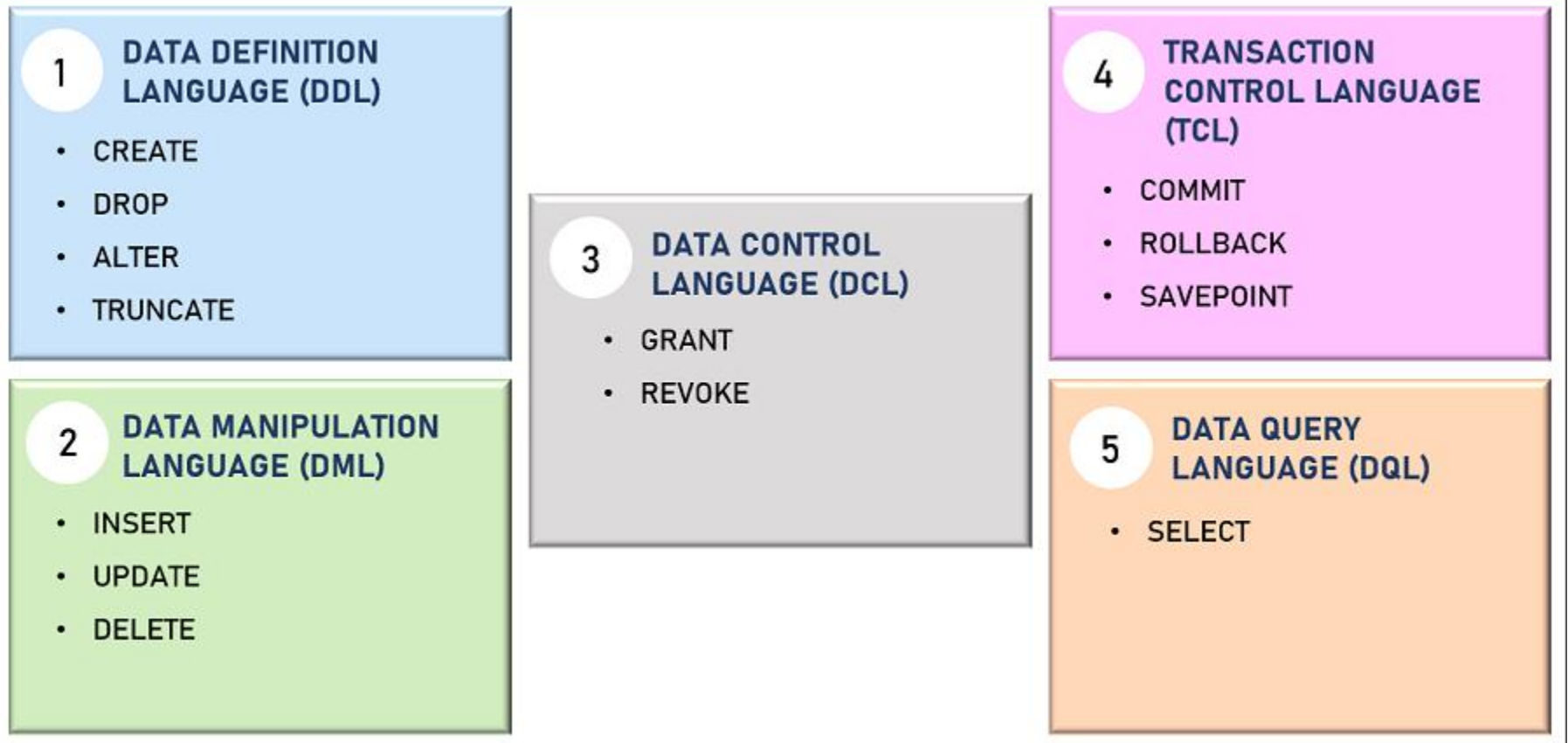
Datetime Data Types

- You can use several datetime data types:

Data Type	Description
TIMESTAMP	Date with fractional seconds
INTERVAL YEAR TO MONTH	Stored as an interval of years and months
INTERVAL DAY TO SECOND	Stored as an interval of days, hours, minutes, and seconds



SQL Commands



Common SQL Functions

- **Aggregate Functions:** COUNT(), SUM(), AVG(), MIN(), MAX()
- **String Functions:** UPPER(), LOWER(), CONCAT(), SUBSTRING()
- **Date Functions:** NOW(), CURDATE(), DATEDIFF()

SQL command exercises

1. Create a Table: Create a table Students with the following columns:

- StudentID → integer, primary key
- FirstName → varchar(50)
- LastName → varchar(50)
- Age → integer
- Major → varchar(50)

2. Insert Data into a Table: Insert the following students into the table:

StudentID	FirstName	LastName	Age	Major
1	Ali	Omar	20	Computer
2	Sara	Ahmed	22	Math
3	Rami	Khalid	21	Physics